LANDSCAPE HEALTH MONITORING

at the Humber Arboretum

DE-BUG GANG

Project Plan - Spring 2026

Armend • Adrian • Mats • Aidan • Mayca

Table of Contents

1. Introduction....3
2. Problem Statement Definition....3
2.1 Humber Office of Sustainability....3
2.2 Public Visitors....4
3. 'How Might We' Questions....5
4. Chosen 'How Might We' Questions....5
5. Success Criteria....6
5.1 High Priority....6
5.2 Standard Priority....6
6. Stakeholder Analysis....7
6.1 Primary Stakeholder - Humber Office of Sustainability....7
6.2 Secondary Stakeholder - Public Visitors....7
7. Design Principles....8
8. Project Timeline & Milestones....9
9. Risk, Constraints & Mitigations....12
9.1 Constraints....12

1. Introduction

The Humber Arboretum is a public garden and ecological landscape located within the Humber River watershed in Toronto. The arboretum functions as a space for biodiversity conservation, environmental education, and sustainability initiatives. Because of its ecological value and educational role, it requires effective monitoring of landscape elements such as vegetation, biodiversity, and environmental conditions.

Several stakeholders are involved in managing and interacting with the arboretum, including the **Humber Office of Sustainability, arboretum staff, conservation groups, students, public visitors etc**. However, collecting, organizing, and analyzing landscape data can be challenging due to fragmented monitoring systems and limited accessibility of environmental information.

The goal of this project is to explore how a **digital solution** could support stakeholders in **monitoring landscape health, visualizing ecological data, and improving sustainability reporting**, while also increasing community engagement with the arboretum environment. We have 2 stakeholders that we picked. Our primary stakeholder being the **Humber Office of Sustainability and public visitors.**

2. Problem Statement Definition

2.1 Humber Office of Sustainability

Problem Statement

The Humber Office of Sustainability needs a way to systematically monitor, collect, record, and analyze data on landscape elements across the Humber Arboretum and surrounding campus properties, because they currently lack a centralized and reliable system to track ecological metrics and landscape features such as green space coverage.

This is important because Humber College has committed to achieving Net-Zero emissions by 2029. Without accurate, ongoing, and reportable landscape data, the Office cannot effectively:

- Measure progress against Net-Zero targets
- Identify gaps in sustainability initiatives
- Make evidence-based land management decisions
- Produce credible sustainability reports for institutional leadership, funders, and the broader community

2.2 Public Visitors

Problem Statement

Public visitors to the Humber Arboretum need a way to access meaningful, easy-to-understand information about the ecological features and sustainability efforts across the Arboretum grounds, because they currently have limited visibility into landscape elements such as tree species, biodiversity, and the College's broader environmental initiatives.

This is important because without an engaging and informative visitor experience:

- Opportunities to build public environmental awareness are lost
- Community stewardship and broader support for sustainability cannot be effectively fostered
- The arboretum's impact is limited to its ecological function alone
- Support for hobbyists and citizen scientists is constrained, reducing the platform's value for non-technical users

3. ‘How Might We’ Questions

The following ‘How Might We’ (HMW) questions are intended to reframe the problem space and explore opportunity areas:

- How might we make contributing landscape data feel straightforward for both trained arboretum staff and everyday visitors, so neither group is left out of the process?
- How might we present complex metrics in a way that a public visitor can quickly grasp, without watering down the science behind it?
- How might we build a data reporting system that automatically translates landscape health data into ESG-compliant outputs, reducing manual work for the Office of Sustainability?
- How might we give the Office of Sustainability a reliable way to monitor green space vs. hardscape changes and demonstrate measurable progress toward the Net-Zero 2029 target?
- How might we design map visualizations that let any user explore landscape elements and their ecological data freely, with minimal technical knowledge required?
- How might we build the data architecture to be modular enough that administrators can introduce new landscape element types as monitoring requirements grow?
- How might we open the door to public visitors and citizen scientists as meaningful contributors, while keeping the data quality high enough for institutional reporting?

4. Chosen ‘How Might We’ Questions

Selected HMW Question

How might we design a data reporting system that automatically translates landscape health metrics into ESG-compliant sustainability reports for the Office of Sustainability?

This question was chosen because it speaks directly to what the primary stakeholder needs most right now: credible, auditable sustainability reporting that holds up against Humber College's 2029 Net-Zero commitment. Getting this right also lays the data foundation that secondary stakeholders, like public visitors and citizen scientists, can eventually build on.

The other HMW questions are still valuable, but things like public engagement, map visualizations, and modular data entry are best thought of as supporting features. They feed into and strengthen the core reporting and monitoring system that this question is really about.

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5. Success Criteria

The proposed solution should satisfy the following criteria, organized by priority:

5.1 High Priority

These features directly support the core monitoring and reporting needs of the Office of Sustainability:

Success Criterion	Rationale
Visual dashboards for monitoring landscape health metrics	Enables at-a-glance monitoring of ecological KPIs by sustainability staff
Accurate environmental data with ≥95% reliability	Ensures institutional reporting is credible and auditable
Interactive map visualization of hardscape and softscape areas	Supports spatial understanding of green space coverage across the campus
Expandable landscape element information panels	Allows users to explore detailed ecological data without leaving the map
Validated data entry and import functionality	Prevents data integrity issues when ingesting field observations or sensor data

5.2 Standard Priority

These features enhance usability and expand the platform's reach:

- Automatic generation of sustainability reports aligned with ESG guidelines and STAR Platinum rating standards
- Community interaction and citizen science data contribution features
- Staff update section for sharing sustainability news and initiatives
- Percentage overview of hardscape vs. softscape coverage
- Budget-friendly architecture and sustainable long-term maintenance model

6. Stakeholder Analysis

6.1 Primary Stakeholder - Humber Office of Sustainability

Role	Responsible for monitoring sustainability metrics and reporting environmental performance across Humber College campus properties.
Goals	Achieve Net-Zero emissions by 2029; produce credible ESG and STAR sustainability reports; systematically track landscape health.
Pain Points	Lack of a centralized system; fragmented data sources; manual and time-consuming reporting processes; difficulty demonstrating measurable progress.
Needs from Solution	Reliable data collection; automated report generation; landscape map with historical tracking; validated import workflows.
Influence	High – primary decision-maker and end user for institutional features of the platform.

6.2 Secondary Stakeholder - Public Visitors

Role	Visitors of the Humber Arboretum who explore the grounds for recreation, education, or personal interest in nature and sustainability.
Goals	Learn about plant species, wildlife, and sustainability initiatives; engage meaningfully with the ecological environment around them.
Pain Points	Limited access to information about what they are observing; no way to contribute to the sustainability story of the space; limited support for hobbyist or citizen science use cases.
Needs from Solution	Simple, approachable interface; species and ecological information accessible on mobile; ability to submit observations; engaging sustainability updates.
Influence	Medium – engagement and community contributions help validate and enrich the platform's data layer.

7. Design Principles

The following principles guide all design and development decisions for this project:

Accessibility First

The platform must be usable by visitors with no technical or scientific background, as well as by sustainability professionals. All features should be discoverable without training.

Data Integrity

All data collected - whether from sensors, manual entry, or community contributions, must pass validation before being stored or included in reports. Accuracy is non-negotiable for institutional reporting.

Transparency

Ecological metrics, sustainability progress, and data sources should be clearly communicated to all user types. The platform should never obscure uncertainty or data gaps.

Modularity

The system architecture should allow administrators to add new landscape element types, monitoring metrics, and data sources as the arboretum's needs evolve over time.

Community Empowerment

Public visitors and citizen scientists should feel that their contributions are valued and have real impact. Contribution pathways should be simple and rewarding.

Sustainability by Design

The platform itself should be lightweight, cost-effective, and maintainable long-term - reflecting the sustainability values it helps to track and report.

8. Project Timeline & Milestones

The table below shows the full 16-week project schedule across Spring 2026, mapped to course modules and key milestones. Project focus in each week aligns with the active module.

Course Module	Presentations	Online Week	Travel	Local Holiday	Reparation Wks

WK	DATE	MON	TUE	WED	THU	FRI	PROJECT FOCUS
1	23/02–01/03	Online Week				Travel to Eindhoven	Introductory
2	02/03–08/03	Human Centered Design – Empathize: brand audit, stakeholder interviews					Empathize & Define phase
3	09/03–15/03	Human Centered Design – Empathize: competitor research, user survey					Ideate phase
4	16/03–22/03	Human Centered Design – Define: problem statement, HMW questions, personas					UX design delivery
5	23/03–29/03	Human Centered Design – Define: success criteria, design principles			Presentations	Travel to Austria	Build testable prototypes
6	30/03–05/04	Online Week				Travel to Austria	
7	06/04–12/04	Local holiday	Software Engineering – Ideate: moodboa				Requirements Analysis Design Patterns

			rds, wireframes, architecture				
8	**13/04–19/04**	Software Engineering – Ideate: dashboard concepts, data entry flows					Web UI Backend Logic
9	**20/04–26/04**	Software Engineering - Prototype: interactive prototype, usability plan					Data Persistence REST APIs
10	**27/04–03/05**	Software Engineering - Prototype: ESG report generator, map overlay			**Presentations**	**Travel to Chania**	DevOps Automation Code Quality
11	**04/05–10/05**	IoT – Test: usability testing with stakeholders, feedback sessions					Supervision of IoT Projects SCRUM Methodology
12	**11/05–17/05**	IoT – Test: synthesize feedback, iterate on prototype					Key Soft Skills IoT Technology Stack
13	**18/05–24/05**	IoT – Refine: citizen science features, community data contribution flow			**Presentations**	**Travel to Toronto**	Business Modelling for IoT
14	**25/05–31/05**	AI – Refine: map visualizat					AI/ML concepts

		ion, hardscape/softscape overlays					
15	**01/06–07/06**	AI – Finalize: polish prototype, complete documentation					AI/ML concepts
16	**08/06–14/06**	AI – Final deliverables: complete project plan, prototype & handover			**Presentations**	**Travel home**	AI/ML concepts
19–20	**15/06–28/06**	**Reparation Weeks**					

Note: All project deliverables should be finalized by the end of Week 16 (14 June 2026). The team will hold weekly check-ins to assess progress and adapt focus as needed.

9. Risk, Constraints & Mitigations

The following table identifies key risks, their likelihood and impact, and the team's planned mitigations:

No.	Risk	Likelihood	Impact	Mitigation Strategy
1	Inconsistent and outdated landscape data	High	High	Implement data validation rules at point of entry; establish data review schedule with sustainability staff
2	Unknown hardware availability and quality for sensors or IoT integration	High	Low	Design solution to function with manual data entry as baseline; treat sensor integration as an enhancement
3	Budget constraints limiting technology choices	Med	Low	Prioritize open-source tools and cloud-free or low-cost hosting options; scope features to match available resources
4	System input validation failure allowing corrupt data into reports	High	Med	Build multi-layer validation (client-side + server-side); flag anomalies for staff review before reporting
5	Data privacy concerns when collecting community observations	Med	High	Anonymize public contributions by default; obtain informed consent; align with FIPPA guidelines and Humber's data policy

9.1 Constraints

In addition to the risks above, the following project constraints have been identified:

- The solution must be maintainable by the Humber Office of Sustainability without dedicated IT support
- All features must be accessible on mobile devices, as public visitors will primarily use smartphones in the field
- Sustainability reporting outputs must align with established frameworks (ESG guidelines, STAR Platinum) to be accepted by institutional stakeholders
- Community-contributed data must be clearly distinguished from verified institutional data within all reports
- The platform should be deployable within the Spring 2026 semester timeline